

Reconnaissance (Information Gathering)

Reconnaissance is the first phase of ethical hacking and penetration testing. It involves collecting information about a target system, network, or organization before attempting any security testing.

1. Passive Reconnaissance

Passive Reconnaissance is the process of gathering information without directly interacting with the target system. The target is usually unaware that information is being collected.

A) Whois:

Whois is a protocol used to obtain information about a domain name. Information gathered includes domain owner details, registrar information, registration and expiry dates, and name servers.

Example: `whois example.com`. It helps identify the organization behind a website.

B) Nslookup:

Nslookup is a command-line tool used to query DNS records. It provides IP addresses, DNS records, and mail server information.

Example: `nslookup example.com`. It maps a domain name to its IP address.

C) Google Dorking:

Google Dorking uses advanced search operators to find publicly available information.

Examples include `site:example.com`, `intitle:'index of'`, and `filetype:pdf site:example.com`. It helps identify exposed documents and web pages.

D) Shodan:

Shodan is a search engine for internet-connected devices. It provides information about open ports, services, device types, and server banners. It helps discover exposed systems on the internet.

2. Active Reconnaissance

Active Reconnaissance involves direct interaction with the target system. The target may detect these activities through logs or monitoring tools.

A) Ping Sweep:

A Ping Sweep is used to identify active hosts on a network.

Example: `nmap -sn 192.168.1.0/24`. It helps find live systems and active IP addresses.

B) Banner Grabbing:

Banner Grabbing collects information about services running on a target.

Example: `nmap -sV target_ip`. It reveals service names, software versions, and operating system details.